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**B.TECH. IV SEM MAIN EXAM
AUGUST-2023**

**COMPUTER SCIENCE AND ENGINEERING
(4CS1-01) - DISCRETE MATHEMATICS
COMMON TO CS, IT, AI, DS, MC, CM, CD, CA,
AD, AM, CY, IO**

Time : 3 Hours]

[Max. Marks : 70

[Min Passing Marks : 23]

Instructions to Candidates : Part – A : Short answer type questions (up to 25 words)

10 × 2 marks = 20 marks. All ten questions are compulsory.

Part – B: Analytical/Problem Solving questions 5 × 4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part – C: Descriptive/Analytical/Problem Solving questions 3 × 10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1 _____

2 _____

PART A

Q. 1. How many one-to-one functions are there from a set with five elements to a set with 10 elements ?

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- Q. 2/ What is a bipartite graph ? Give example.
- Q. 4/ Simplify the expression “(P or Q) and (not P)” using logical operation.
- Q. 4/ Define homomorphism and isomorphism of groups X
- Q. 5/ What is a generating function ? Give example.
- Q. 6/ What is the difference between path and circuit ? Define hamiltonian path and circuit.
- Q. 7/ Define the predicates and quantifiers in propositional logic.
- Q. 8/ Define the properties of equivalence relation.
- Q. 9/ What is Pigeonhole Principle ?
- Q. 10/ Define the symmetric difference of two sets using venn diagram.

PART B

- Q. 1/ What is the minimum number of students required in a discrete mathematics class to be sure that atleast six will receive the same grade, if there are five possible grades, A, B, C, D and F.
- Q. 2. Suppose that number of bacteria in a colony triples every hour :
- (i) Set up a recurrence relation for the number of bacteria after “n” hours have elapsed.
 - (ii) If 100 bacteria are used to begin a new colony, how many bacteria will be in the colony in 10 hours.
- Q. 3. Give the accursive definition a function “L” from the set “s” of strings over $\{a, b\}$ to the set of natural numbers that gives the length of a string.
- Q. 4. / Prove by contradiction that $\sqrt{2}$ is an irrational number.
- Q. 5/ Given sets $A = \{1, 2, 3\}$ and $B = \{2, 5, 4\}$, find union, intersection and implements of set A and B.

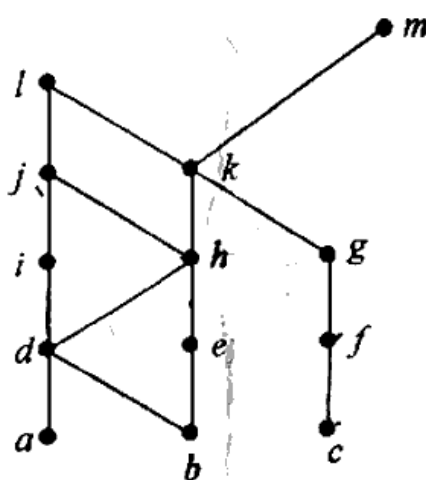
Q. 6. Determine if the set $\{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$ forms a lattice under the relation "divides".

Q. 7. How many ways are there to arrange the letters in the word "COMBINATIONS" such that all vowels are together?

PART C

Q. 1. Find the generating function for the sequence $\{1, 1, 2, 3, 5, 8, \dots\}$ defined by fibonacci numbers.

Q. 2. Answer these questions for the partial order represented by following Hasse diagram.



- Find the maximal and minimal elements.
- Is there a greatest element? Is there a least element?
- Find all upper bounds of $\{a, b, c\}$.
- Find least upper bound of $\{a, b, c\}$, if it exists,
- find all lower bounds of $\{f, g, h\}$.
- Find greatest lower bound of $\{f, g, h\}$, if it exists.

Q. 3. Show that the function $f(x) = |x|$ from the set of real numbers to the set of non-negative real numbers is not invertible but if the domain is restricted to the set of non-negative real numbers, the resulting function is invertible.

